IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE PART II: MEng, BEng and ACGI

DEVICES

Tuesday, 26 May 10:00 am

Time allowed: 1:30 hours

There are TWO questions on this paper.

Answer ALL questions. Question One carries 25 marks and Question Two carries 25 marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : W.T. Pike
Second Marker(s) : Z. Durrani

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Special instructions for invigilators

Special instructions for students

Do not use red nor green ink.

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Constants

permittivity of free space:	$\varepsilon_o = 8.85 \times 10^{-12}$ F/m
permeability of free space:	$\mu_o = 4\pi \times 10^{-7}$ H/m
intrinsic carrier concentration in Si:	$n_i = 1.45 \times 10^{10}$ cm^{-3} at $T = 300$ K
dielectric constant of Si:	$\varepsilon_{Si} = 11$
dielectric constant of SiO_2:	$\varepsilon_{ox} = 4$
thermal voltage:	$V_T = kT/e = 0.026$ V at $T = 300$ K
charge of an electron:	$e = 1.6 \times 10^{-19}$ C
Planck's constant:	$h = 6.63 \times 10^{-34}$ Js
bandgap of Si:	$E_G = 1.12$ eV at $T = 300$ K
electron affinity of Si	$\chi = 4.05$ eV at $T = 300$ K
effective density of states of Si:	$N_C = 3.2 \times 10^{19}$ cm^{-3} at $T = 300$ K $N_V = 1.8 \times 10^{19}$ cm^{-3} at $T = 300$ K

Formulae

$$\left.\begin{aligned} J_n(x) &= e\mu_n n(x)E(x) + eD_n \frac{dn(x)}{dx} \\ J_p(x) &= e\mu_p p(x)E(x) - eD_p \frac{dp(x)}{dx} \end{aligned}\right\}$$

Drift-diffusion current equations

$$\left.\begin{aligned} \frac{\partial \delta n}{\partial t} &= D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n} \\ \frac{\partial \delta p}{\partial t} &= D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p} \end{aligned}\right\}$$

Continuity equations of minority carriers

$$\left.\begin{aligned} J_n &= \frac{eD_n n_p}{L_n}\left(e^{\frac{eV}{kT}} - 1\right) \\ J_p &= \frac{eD_p p_n}{L_p}\left(e^{\frac{eV}{kT}} - 1\right) \end{aligned}\right\}$$

Text-book diode diffusion currents

$$V_0 = \frac{kT}{e}\ln\left(\frac{N_A N_D}{n_i^2}\right)$$

Built-in voltage

$$c = c_0 \exp\left(\frac{eV}{kT}\right) \text{ with } \begin{cases} c = p_n \text{ or } n_p \\ c_0 \text{ bulk minority carrier concentration} \end{cases}$$

Minority carrier injection under bias *V*

$$\delta c = \Delta c \exp\left(\frac{-x}{L}\right) \text{ with } \begin{cases} \delta c = \delta p_n \text{ or } \delta n_p \\ \Delta c \text{ the excess carrier concentration} \\ \text{at the edge of the depletion region} \end{cases}$$

Excess carrier concentration as a function of distance when recombination occurs – long layer approximation.

$$L = \sqrt{D\tau}$$

Diffusion length

$$D = \frac{kT}{e}\mu$$

Einstein relation

$$C_{diff} = \frac{e}{kT} I\tau$$

Diffusion capacitance

$$i(t) = \frac{Q(t)}{\tau} + \frac{dQ(t)}{dt}$$

Time variation of current and charge

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon}$$

Poisson equation in 1 dimension

$$n = N_C \, exp\left(\frac{E_F - E_C}{kT}\right)$$

$$p = N_V \, exp\left(\frac{E_V - E_F}{kT}\right)$$

Carrier concentrations

1. Long vs short diodes

a) Explain the process by which minority carriers can move from the junction region of a diode to the contacts under forward bias. Define in terms of the relevant parameters of the carriers in the diode what is meant by the long- and short-diode approximation. [4]

b) A p^+n diode has p doping of 10^{16} cm^{-3} and n of 10^{14} cm^{-3}. Draw two diagrams showing the variation of the excess electron and hole minority carrier densities along the entire length of a diode under forward and reverse bias, assuming neither the short- or long-diode approximations apply. For a forward and reverse bias of 1V on this diode, calculate the values of the excess minority hole carrier concentrations at the edges of the depletion region and at the contacts. [7]

c) The steady-state current flow in a diode can be derived by looking either at the current flow at the edge of the depletion region or the charge needed to be resupplied to maintain a steady state. From equating these two approaches show that the excess minority carrier concentration, δp_n in the *n* region of the diode a distance x from the depletion region will be given by solving

$$\delta p_n = D_p \tau_p \frac{d^2 \delta p_n}{dx^2}$$

where D_p is the diffusion constant and τ_p the recombination time. Solve this equation using appropriate boundary conditions for the long-diode approximation.

Why is it not possible to use this equation to solve for the short-diode approximation? [14]

2. Switching delays

a) What is the dominant recombination process of carriers in silicon semiconductors under low electric fields, and why are other recombination processes slower? How does a high field contribute additional carrier recombination? [6]

b) The expression for the time dependence of the excess hole charge, $Q_p(t)$ is given by

$$\frac{dQ_p(t)}{dt} = i_p - \frac{Q_p(t)}{\tau_p} \qquad 2.1$$

where τ_p is the hole recombination time and i_p is the hole current. A switching test is performed on a p^+n diode in series with a resistive load. In the first test the diode is switched off by removing the forward bias. In the second test the same forward bias is reversed at the time of switching. Assuming expression 2.1, sketch the time variation of the excess hole charge after the switching occurs for each test. In the second test, the excess hole charge in the diode falls to zero at a time *T* after the bias is reversed. Show that after the bias is removed in the first test, half the initial charge still remains at the same time *T*. [14]

(c) What will cause additional delay in the second test before the current is completely switched off? Sketch the variation of current in a real example of diode switching, noting the cause of any differences from ideal behaviour. [5]